



Assistive Robotics Part 2



Agenda

Recap of Assistive Robotics Part 1

Human muscles and signals

Muscle activation as an input

Workshop Project



Recap of Assistive Robotics Part 1

Let's run it back!

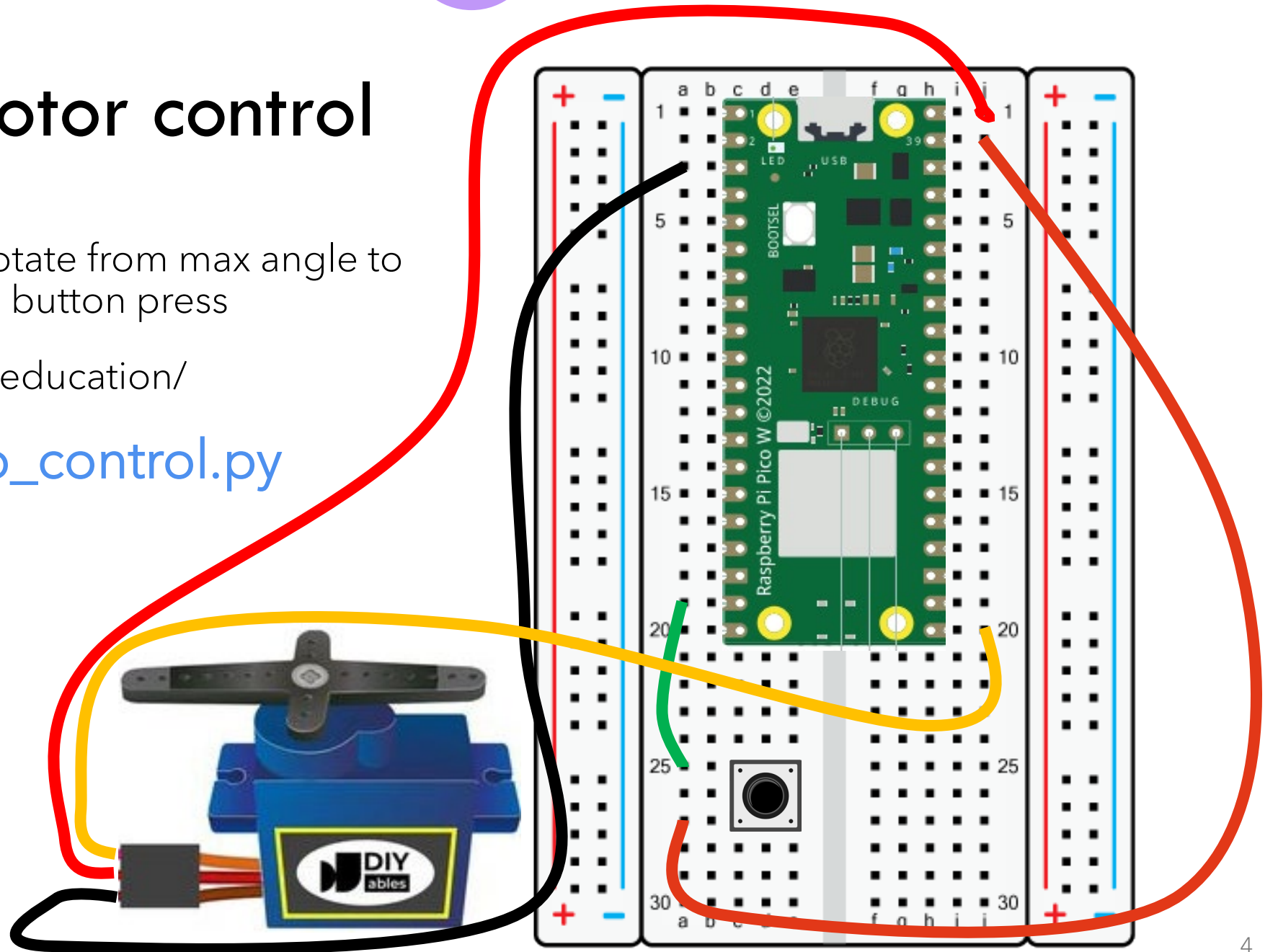
Activity:

Simple motor control

Make the motor rotate from max angle to min angle only on button press

rasr-lab.github.io/education/

[Button_servo_control.py](#)



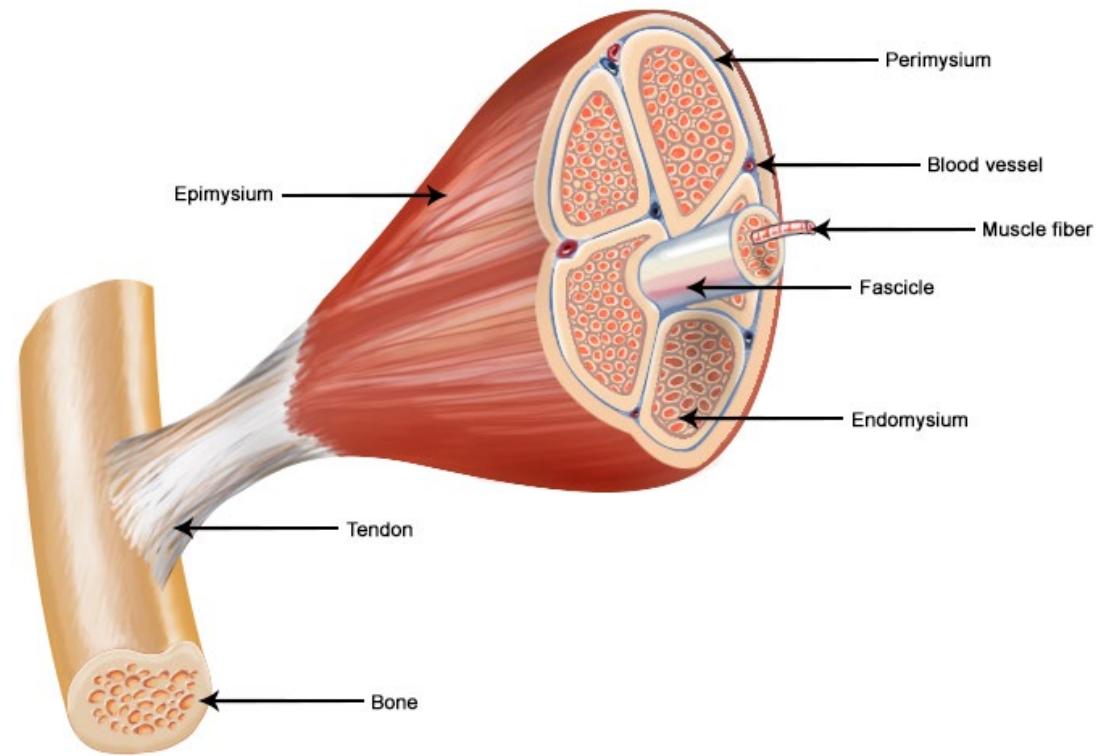


Human muscles and Signals

What makes us move!

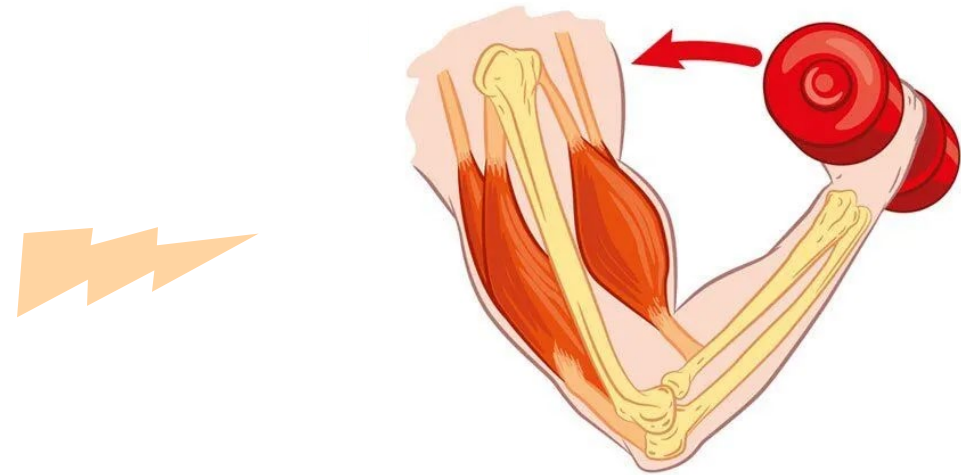
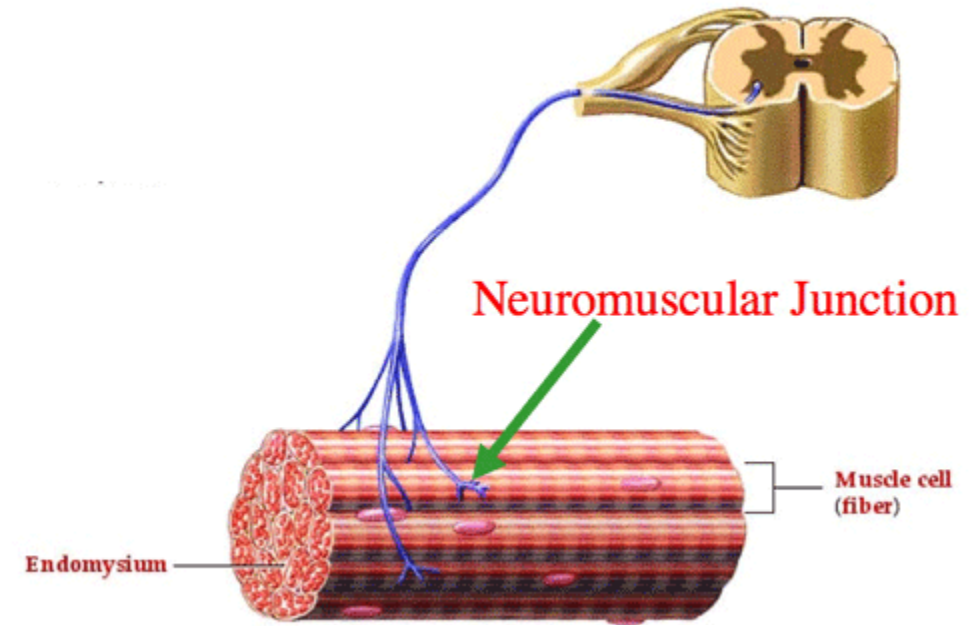
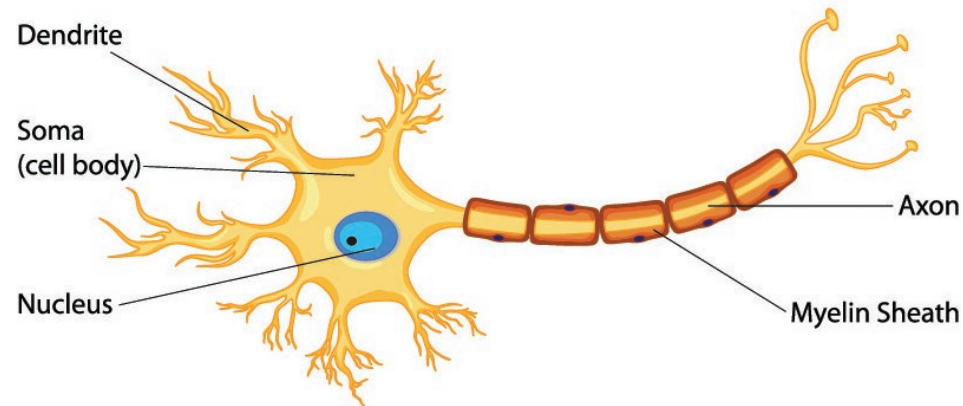
Structure of a muscle

- Soft tissue
- **Muscle fibers** bundled together
- Blood supply within the muscles
- Tendons connect to bones



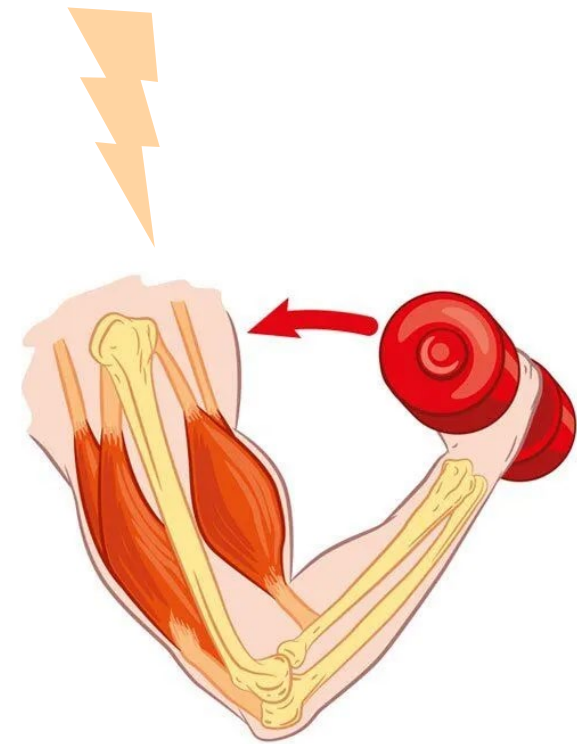
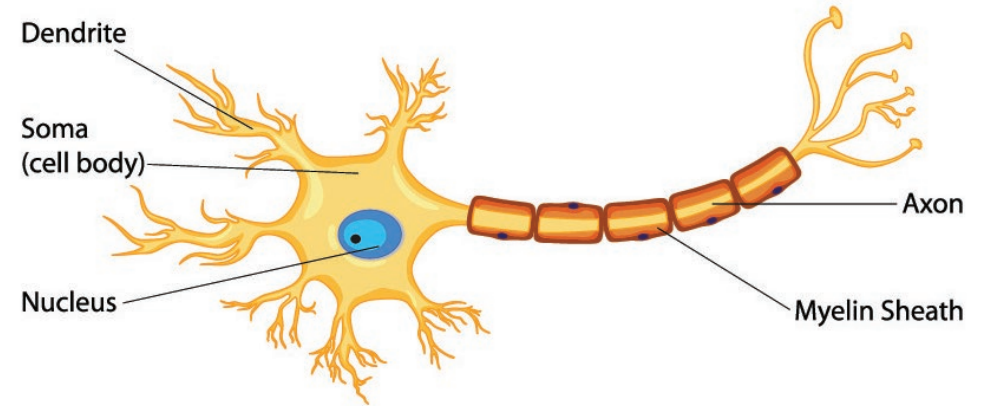
Structure of a muscle

- Muscles need electrical activity to contract
- Nerves supply electrical **stimuli**
- Human nervous system generates electrical stimuli based on intent



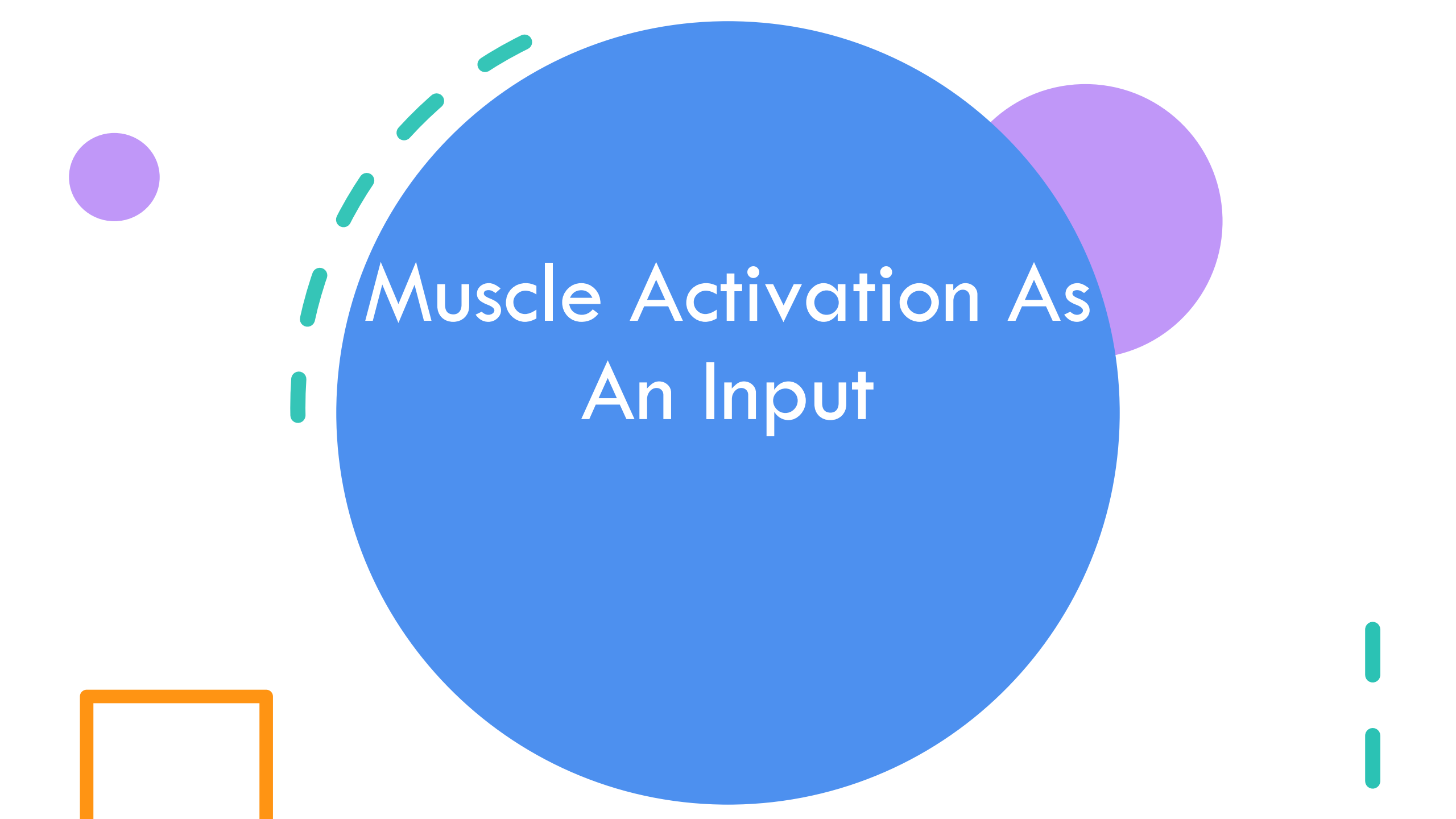
Sensing Muscle Activity

- This “**electrical stimuli**” can be measured
- The tool is called an Electromyograph
 - Needle EMG
 - Fine-wire EMG
 - **Surface EMG**

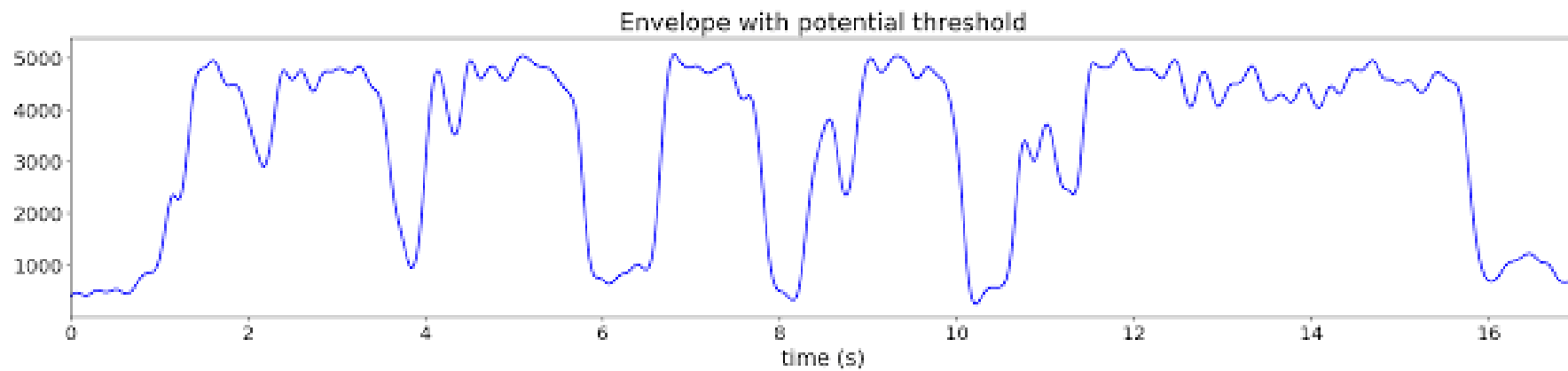
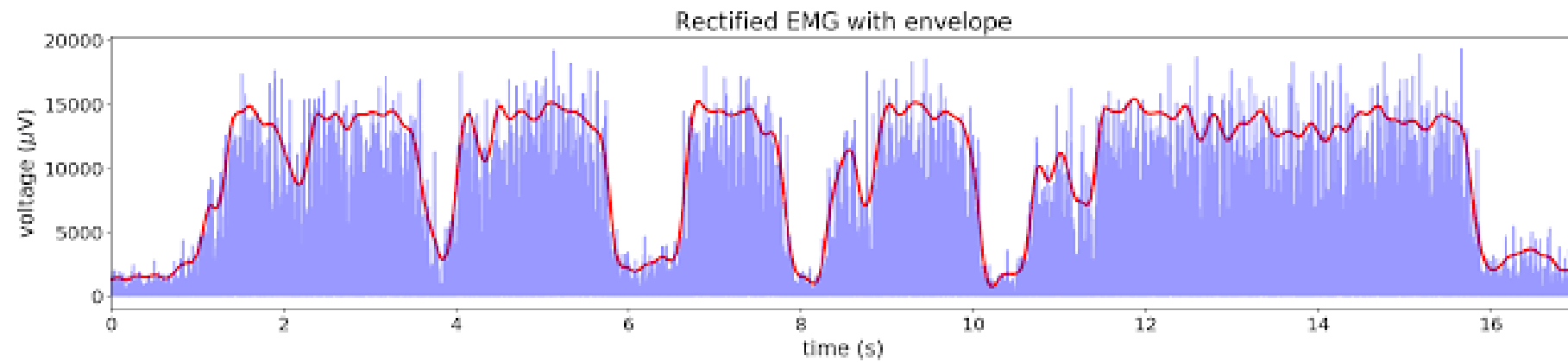
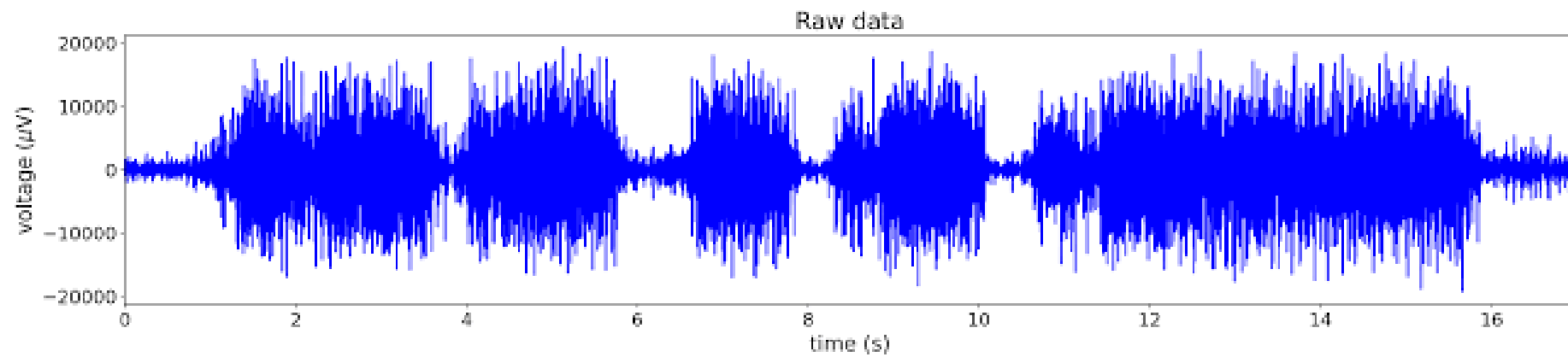


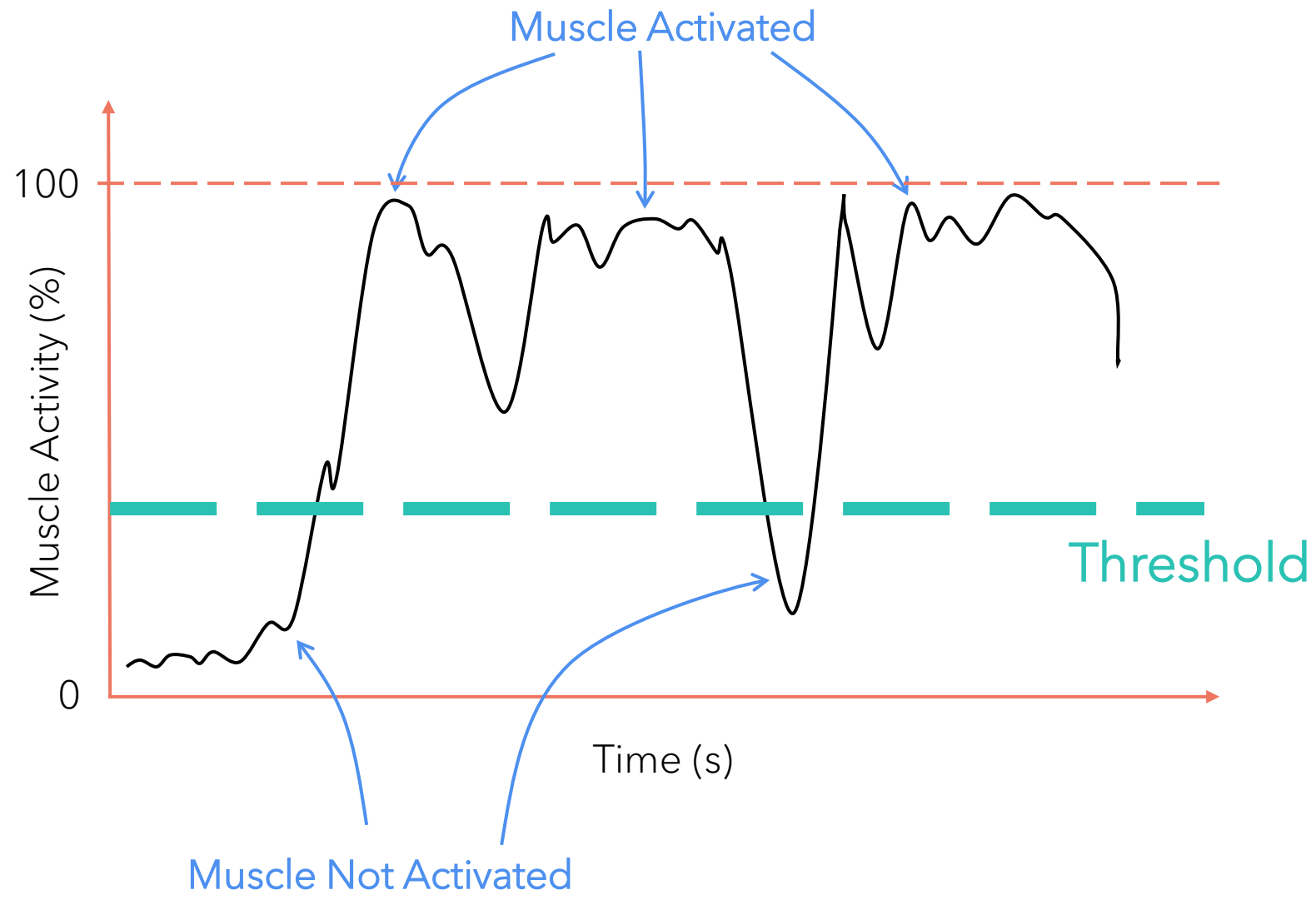


**Let's see an EMG
sensor in action!**



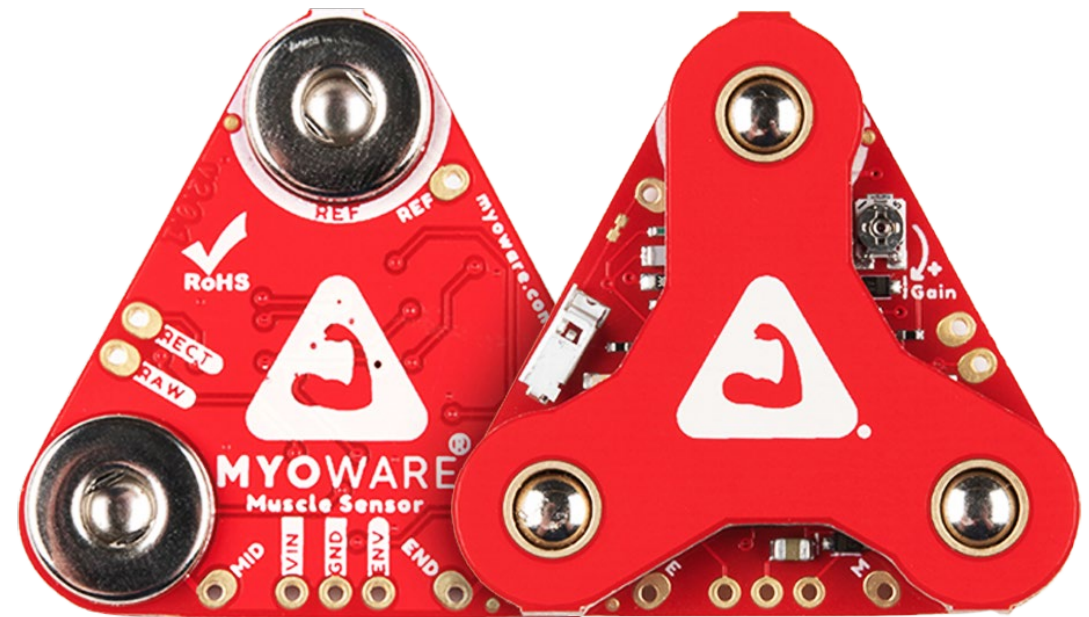
Muscle Activation As An Input





The sensor we will use is

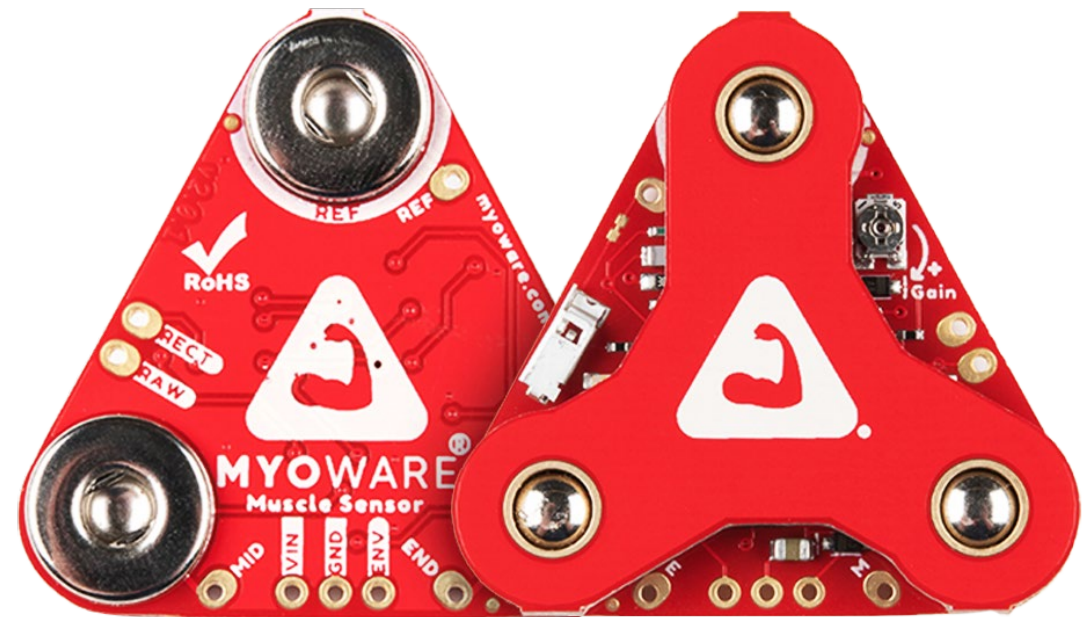
- Can be connected to the microcontroller
- Cheap and convenient
- Readily available
- Already provides an [Envelope Signal](#)



The sensor we will use is

- Red: +5V or Vin
- Black/Green: 0V or GND
- White: ENV (Envelope)

Let's attach the sensor to our biceps



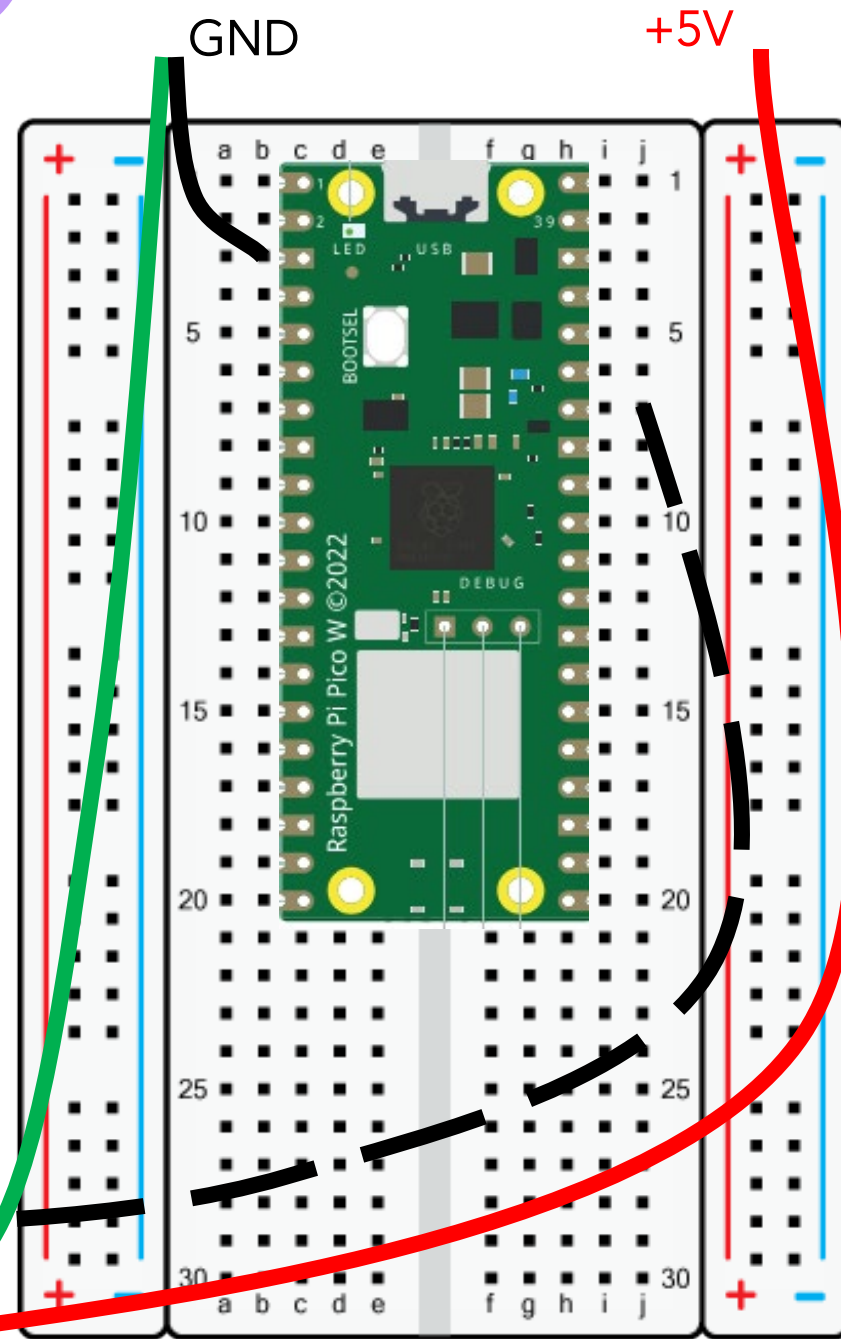
Activity:

Blink the onboard LED

Use the EMG sensor to turn the LED on/off

rasr-lab.github.io/education/

[Blink_LED_EMG.py](#)



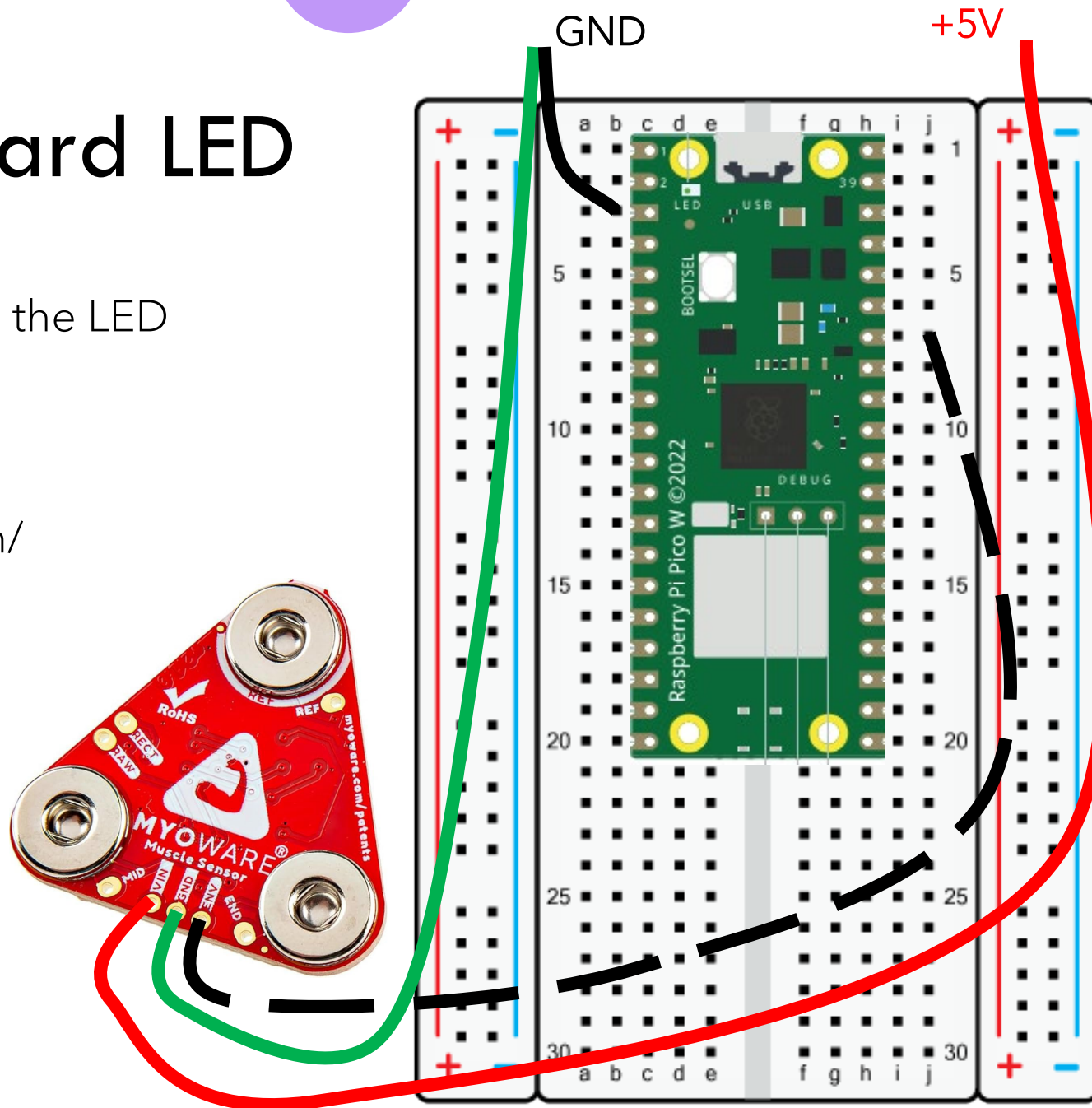
Activity:

Blink the onboard LED

Use the EMG sensor to turn the LED on/off (with a filter)

rasr-lab.github.io/education/

[Blink_LED_EMG_
filtered.py](#)





Workshop Project

Workshop Project

Control an exoskeleton using your muscles

1. Connect the exoskeleton to a microcontroller
2. Attach the exoskeleton to the doll
3. Use your muscle to move the exoskeleton
4. The motor returns to original position once the input goes away

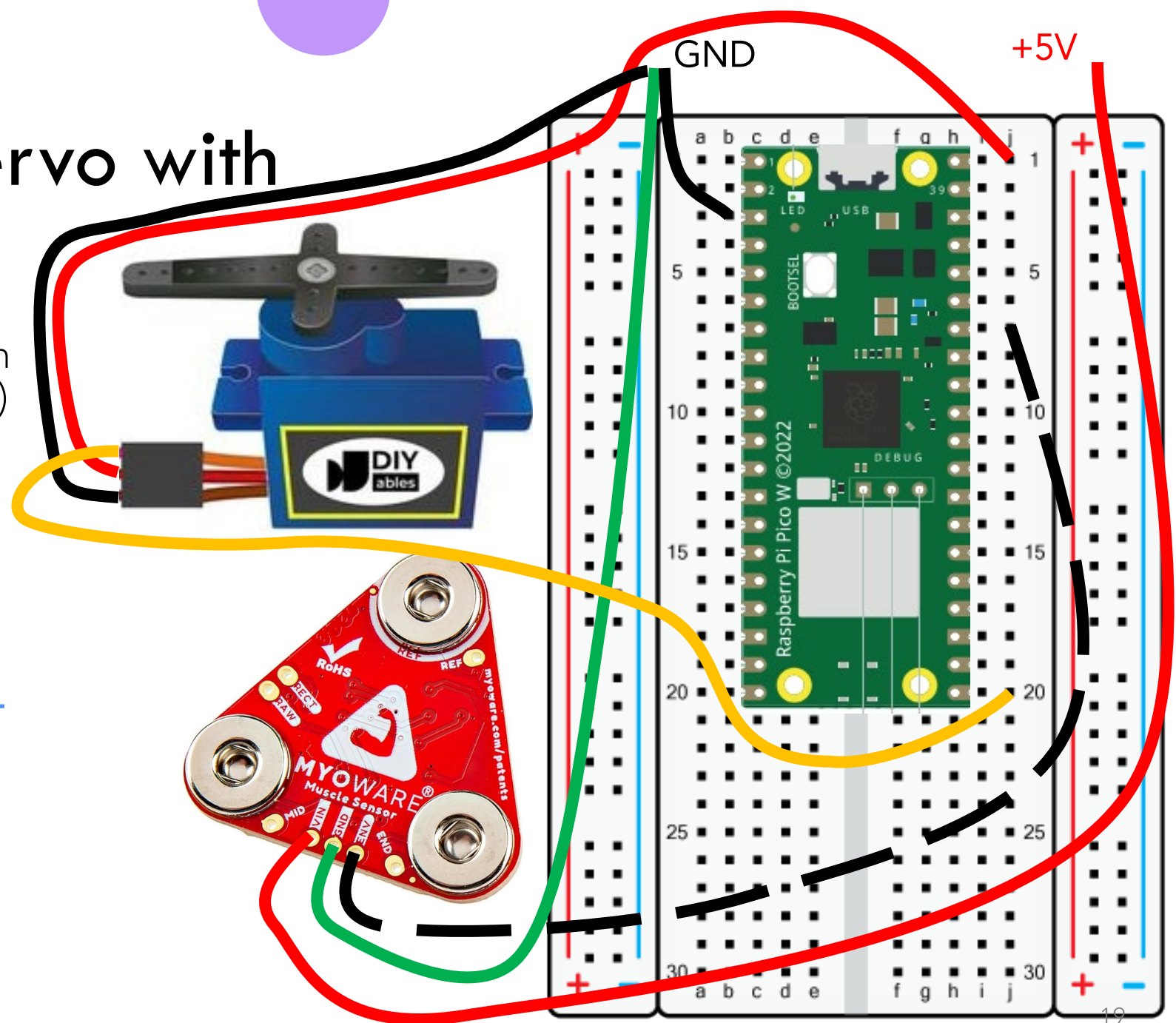


Activity: Control the Servo with your muscle

Use the EMG sensor to turn
the LED on/off (with a filter)

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Exoskeleton_EMG_
Servo.py



Activity:

Attach the Exoskeleton to the Doll

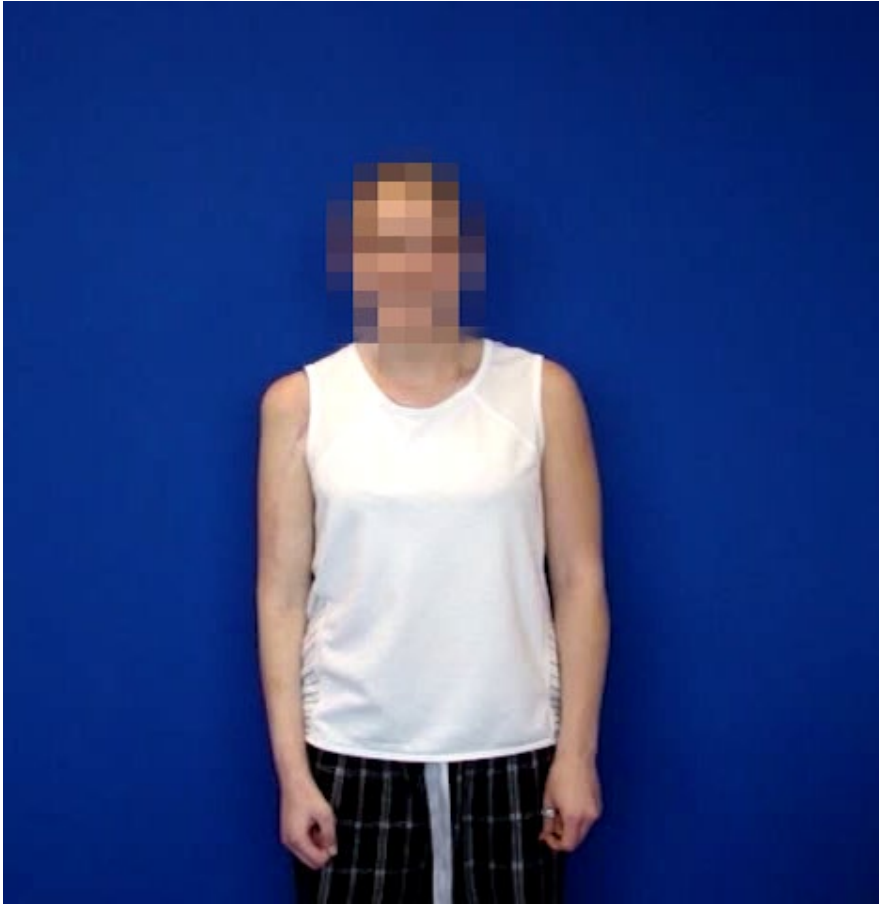
Make sure the Exoskeleton lines up with the elbow joint

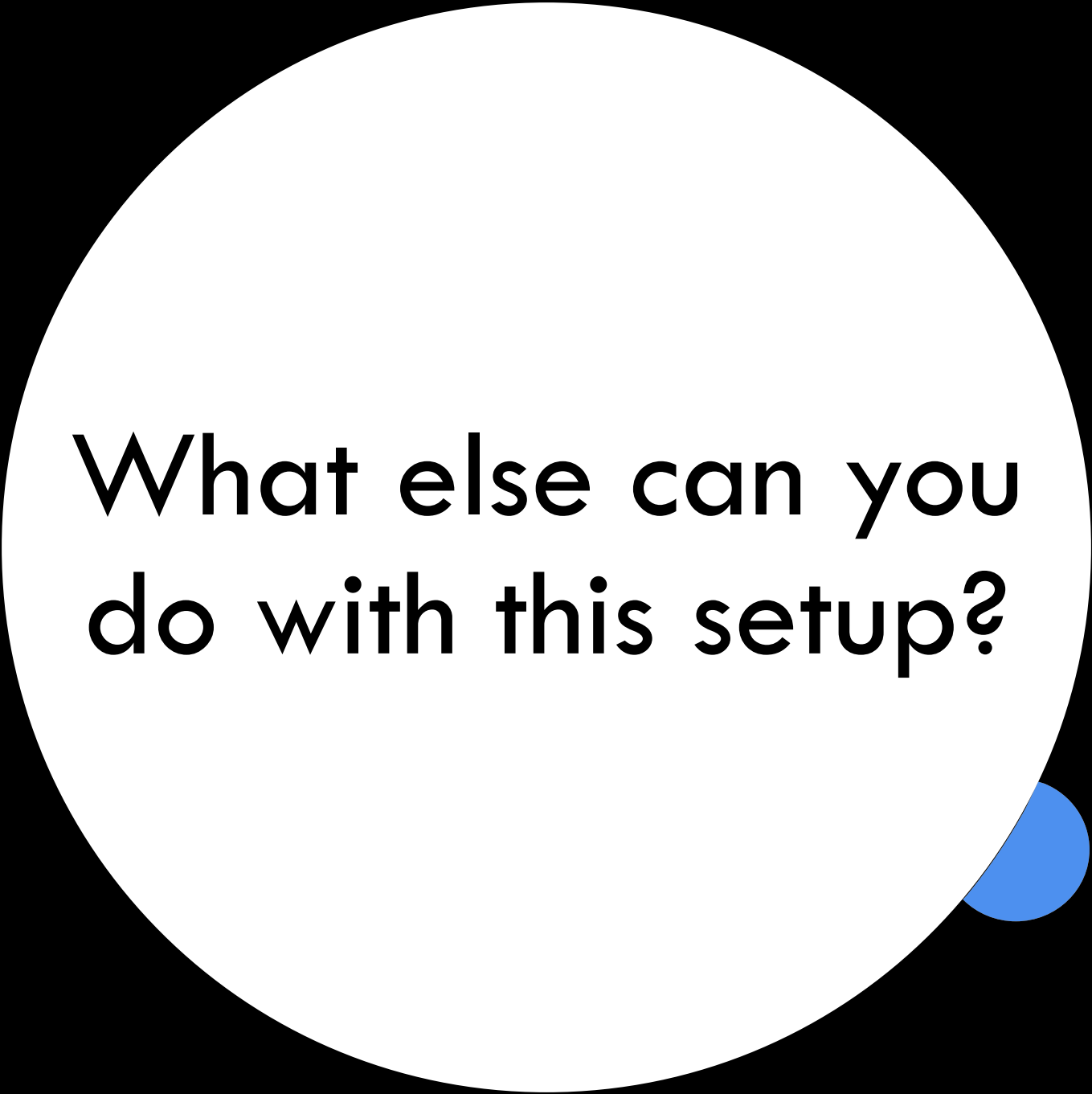
Use masking tape to attach the Exoskeleton to the doll

Make the dolls elbow flex/extend using the Exoskeleton

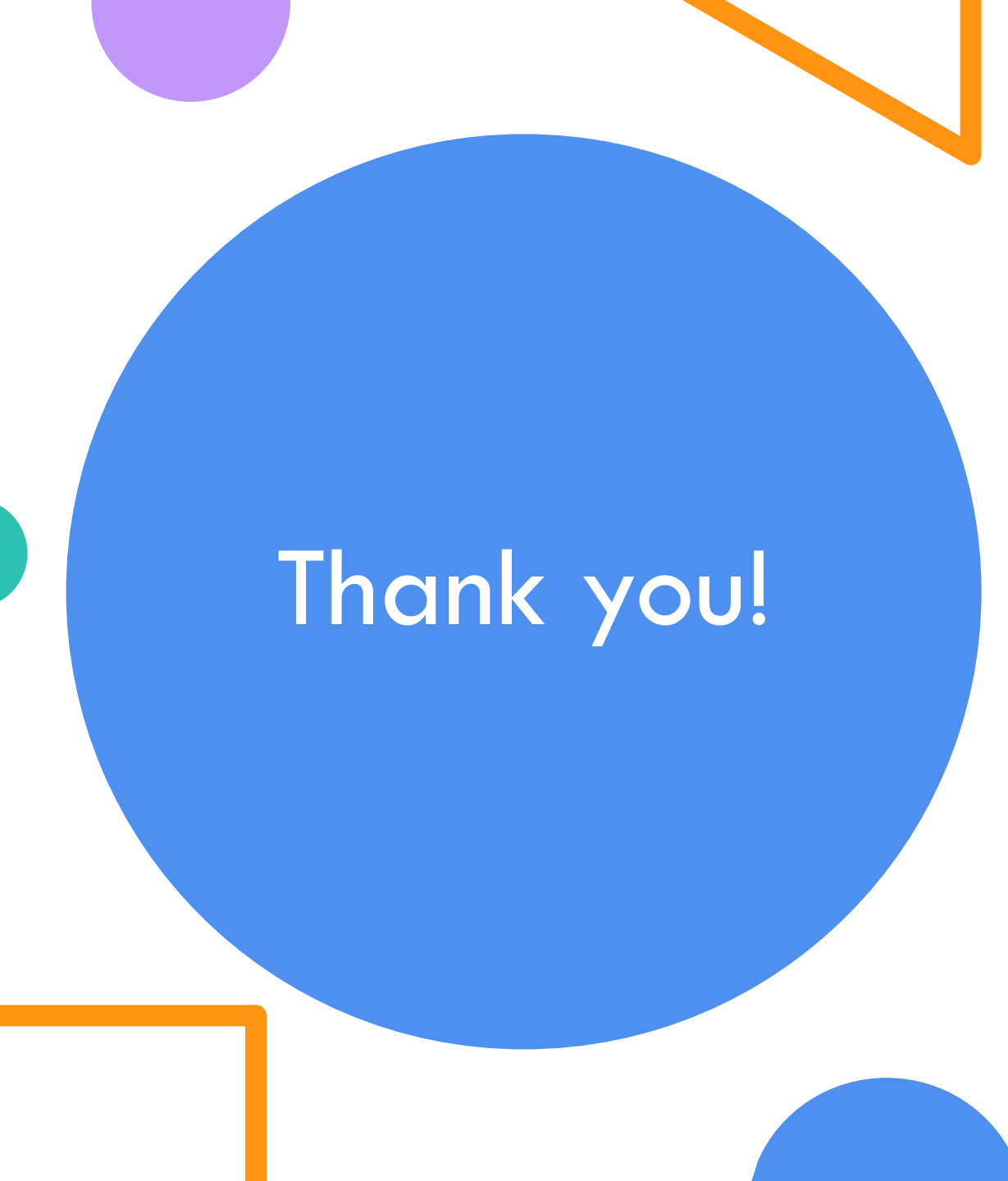


Real World Applications





What else can you
do with this setup?



Thank you!

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